Water Dam Monitoring with Weather Integration

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Abstract. Hydroelectric dams are crucial infrastructures for water supply and electricity generation but face significant operational risks from extreme weather, including flooding during heavy rainfall and reduced energy production during droughts. Current monitoring methods are often manual or outdated, requiring the development of smart, autonomous, and remote solutions. This work presents a Cyber-Physical System (CPS) and Internet of Things (IoT) framework designed for real-time dam monitoring and control. The proposed architecture is mapped to a 5-layer IoT stack and the 5C CPS functional model, using Arduino Nano ESP32 controllers for perception and a Raspberry Pi gateway running Node-RED, RabbitMQ, and InfluxDB. Data management follows an Atomic Messaging philosophy via the MQTT protocol to ensure high scalability and time-series optimization. To enhance operational intelligence, a Gradient Boosting Machine (GBM) model was integrated to predict daily inflow and outflow trends using historical hydrological data and weather forecasts. We implemented a proof-of-concept prototype that successfully validates decision support logic via a real-time Node-RED dashboard. The results provide a robust and secure foundation for adapting integrated CPS technologies to large-scale hydroelectric environments.

Keywords: Cyber-Physical Systems (CPS) · Internet of Things (IoT) · Hydroelectric Dam Monitoring.

1 Introduction

The management of hydroelectric dams represents a critical engineering challenge, as these infrastructures are vital for both sustainable water supply and electricity generation. This project is motivated by the inherent risks associated with dam operations: the danger of overflow during periods of heavy rainfall, which presents significant flooding and safety risks, and the threat of drought during dry seasons, which leads to reduced energy production. Currently, monitoring methods can often be inefficient or outdated. Consequently, there is a clear need for a smart, autonomous, and remote monitoring system to unify and optimize these industrial processes.

The primary objective of this project is to develop a Cyber-Physical System (CPS) and Internet of Things (IoT) framework for dam monitoring and control. The proposed solution collects real-time data on variables such as water level, flow rates and rainfall intensity, supporting compliance with the legally required safety control and observation measures defined for each dam class and associated damage potential, in accordance with the Portuguese dam safety framework established by Decreto-Lei n.º 21/2018 [5], which regulates the safety of large and small dams throughout the phases of design, construction, first impoundment and operation. While the legislation does not prescribe a universal operating range in terms of reservoir percentage, this work adopts a conservative operational strategy in which the CPS helps to keep the reservoir within project-defined safe limits (for example, between 20% and 90% of storage capacity). This is achieved through a distributed architecture using Arduino Nano ESP32 units for local data acquisition and a Raspberry Pi gateway running a Node-RED server and MQTT broker. Furthermore, the system supports predictive decisions using weather forecasts and historical data processed via Gradient Boosting Machine (GBM) models, all visualized through a remote Node-RED dashboard for manual and automatic gate control.

2 Related Work

This section reviews the state-of-the-art in IoT and CPS applied to hydroelectric dams. We also specifically adress architecturees that leverage Machine Learning to enhance monitoring and control mechanisms.

Sadanand et al. [6] present a system integrating **Internet of Things (IoT)** sensors and a **Machine Learning (ML)** model for automated control of dam gates. The system architecture is tripartite: (i) an IoT sensor network strategically deployed in water bodies and within the dam infrastructure; (ii) a predictive model founded on the **XGBoost** algorithm; and (iii) a physical actuation module for gate operation. Water level and velocity data are aggregated and processed by a central unit based on **Raspberry Pi**. The XGBoost model is calibrated to classify hydrological risk into three states (*Alert*, *Warning*, and *Danger*), which subsequently and automatically triggers the progressive opening of the gates.

The authors report an accuracy of 60%, a result obtained through the utilization of a synthetic dataset generated via the **Scikit-learn** library. Notwithstanding its architectural robustness and integrated automation, the system lacks empirical validation with real-world data and presents inherent limitations concerning scalability and reliability under adverse environmental conditions.

Aditya et al. [1] propose a hybrid system that merges IoT technology with **Artificial Neural Networks (ANN)** for monitoring and meteorological forecasting. The model incorporates water level, pressure, and temperature sensors linked to **Arduino Uno** microcontrollers and **ESP8266** modules, leveraging the **ThingSpeak** platform for cloud communication. The collected data are employed in predicting weather conditions and estimating water flow behavior, thereby enabling automated gate actuation in scenarios of imminent risk.

The ANN algorithm demonstrated an accuracy of 87% in predicting meteorological events (i.e., heavy rain, fog, and thunderstorms) following training with environmental data. The architecture is complemented by a mobile application that provides real-time notifications to operators and alerts local populations. Key limitations include the requirement for manual intervention in certain operational phases, the reduced experimental scale (testing limited to two dam prototypes), and the omission of a performance assessment of the model within real operational environments.

Siddula et al. [7] represent one of the initial attempts to apply IoT for real-time dam monitoring. The system utilizes water level sensors connected to microcontrollers and employs Wi-Fi-based wireless communication for data transmission to a central control panel and a mobile application. Upon a rapid increase in water level, the system issues alerts and facilitates the remote control of the gates.

Although this solution offers a low-cost and easily maintainable implementation, its nature is predominantly **reactive**—it does not incorporate data-driven predictive models, relying solely on fixed thresholds for risk detection. Consequently, this approach does not anticipate flood events with sufficient lead time for the deployment of effective preventative measures.

The article published in the journal *Sensors* [3] introduces an advanced methodology for river flow forecasting, based on **Deep Neural Networks** of the **Multi-Layer Perceptron (MLP)** type. The study utilizes historical time series data encompassing flow rate, precipitation, and temperature as input for model training, which aims to predict the volume of water flow over short-term temporal horizons.

The results demonstrate that the MLP model surpasses conventional algorithms, such as linear regression and **Support Vector Machines (SVM)**, in terms of predictive accuracy and stability. The principal contribution of this study lies in validating the applicability of **Deep Learning (DL)** to hydrological forecasting, suggesting that such models can serve as a valuable complement to existing IoT systems. Nevertheless, the investigation is confined to theoretical forecasting and does not address its direct integration with physical gate control systems.

Table 1 summarises the main characteristics, reported performance, and limitations of the reviewed systems.

Reference	**Core Technology Framework**	**AI/ML Model Employed**	**Primary Operational Function**	**Reported Performance Metric**	**Critical Limitations and Gaps**
Sadanand et al. (2021)	IoT + Raspberry Pi Unit	XGBoost (Machine Learning)	Autonomous Sluice Gate Control	Accuracy of 60% (synthetic dataset)	Absence of empirical validation; scalability and reliability constraints in adverse environments.
Aditya et al. (2021)	IoT + ANN + ThingSpeak Platform	Artificial Neural Networks (ANN)	Environmental Monitoring and Meteorological Forecasting	Accuracy of 87% (meteorological prediction)	Manual intervention required; restricted testing scale; no real-world assessment.
Siddula et al. (2018)	IoT + Wi-Fi Communication	None (fixed threshold logic)	Real-Time Monitoring and Alert Generation	Low-cost implementation and ease of maintenance	Reactive nature; no proactive flood anticipation; static risk thresholds.
Sensors (2025)	Deep Learning Methodology	MLP (Deep Learning)	Fluvial Flow Volume Forecasting	Outperforms linear regression and SVM	Theoretical forecasting only; no direct gate control integration.

Table 1. Comparison of related works.

The analysis of the reviewed literature reveals a clear evolution in the application of artificial intelligence to dam monitoring. Early systems, based exclusively on IoT, such as that by Siddula et al. [7], prioritised real-time data collection and visualisation but lacked predictive capability. With the advancement of Machine Learning and Deep Learning models, hybrid solutions emerged, exemplified by the contributions of Sadanand et al. [6] and Aditya et al. [1], capable of forecasting risks and enacting automated responses. Nevertheless, the majority of the proposed systems continue to face significant challenges: namely, the scarcity of real-world data for model training, the inherent limitation of small-scale prototypes, and the absence of robust integration between the predictive modules and the physical control mechanisms. The 2025 study [3] highlights the substantial potential of Deep Learning in accurately forecasting river flow, yet it fails to address the physical execution of the necessary actions in response to the generated alerts. Thus, a discernible gap persists between advanced computational models and their practical implementation within distributed and resilient IoT environments.

In conclusion, the analysed contributions demonstrate substantial progress in the utilisation of IoT, Machine Learning, and Deep Learning for automated dam management. The most recent solutions successfully incorporate predictive models and alert mechanisms; however, the technological maturity remains constrained by factors such as the scarcity of real-time data, the complexity inherent in model calibration, and communication limitations among devices operating in remote environments.

Future research in this domain should, therefore, focus on the following directives:

- Achieving seamless integration between advanced predictive models and the physical control systems.
- Acquiring long-term, real-world datasets for the robust training and validation of predictive models.
- Developing distributed architectures based on **Edge Computing** to enable localised data processing and decision-making.
- Enhancing the resilience and cybersecurity of IoT systems applied to the management of critical infrastructure.

These advancements will facilitate the construction of systems that are more intelligent, scalable, and reliable for disaster prevention and the optimisation of water resource management.

3 Proposed Solution

The proposed solution architecture is a multi-layered Cyber-Physical System (CPS) designed for reliability and scalability. It integrates physical sensing, edge computing, and cloud-based data management to create a resilient monitoring loop for hydroelectric dam operations.

3.1 System Design and Architecture

System Architecture and Layers The system is structured according to the 5-layer IoT architecture, ensuring a clear separation between physical acquisition and business logic:

- **Perception Layer (Smart Connection Level):** This layer consists of Arduino Nano ESP32 controllers interfaced with pressure transducers for level monitoring, water flow sensors, and pluviometers.
- **Network Layer:** Communication is established via MQTT over Wi-Fi for the prototype, utilizing a "Topic Strategy" that prioritizes single-input trunks (e.g., `sensors/#`) to ensure all incoming data is captured automatically.
- **Middleware Layer (Cyber Level):** A Raspberry Pi serves as the central gateway, hosting a RabbitMQ message broker for reliable delivery and an InfluxDB time-series database for high-performance storage.
- **Application and Business Layers:** Node-RED provides the core logic engine and Human-Machine Interface (HMI). It processes data, triggers safety alerts if the dam exceeds safe limits (20%–90%), and presents real-time visualizations for manual override capabilities.

CPS 5C Architecture Mapping The system is further analyzed through the lens of the 5C functional architecture for Cyber-Physical Systems, which facilitates a transition from raw physical data to autonomous configuration.

1. **Connection (Data Acquisition):** This initial layer focuses on the seamless acquisition of data from the physical asset. In our system, this involves the integration of pressure transducers, pluviometers, and power output meters, alongside solenoid valve actuators.
2. **Conversion (Data Preprocessing):** Raw data is processed at the edge to become "smart" and structured. Our implementation utilizes Node-RED for data aggregation, where local pluviometer data is combined with external Weather API streams.
3. **Cyber (Data Processing and Analysis):** This layer creates the "Digital Twin" of the dam. Centralized processing is handled by the RabbitMQ message broker, which ensures data persistence and provides a virtual representation of the system state for historical analysis and storage.

4. **Cognition (Awareness and Insight):** This level applies advanced analytics. We employ Gradient Boosting Machine (GBM) models for AI-based inflow/outflow forecasting, which generates high-level awareness of current conditions to support a Decision Support System (DSS).
5. **Configuration (Self-adaptation and Control):** This final level enables the feedback loop. The system translates AI-driven insights into actual commands (e.g., opening a valve by 15%) while maintaining a "Human-in-the-Loop" through the Node-RED dashboard for manual overrides.

Data and Ingestion Strategy The system employs an "Atomic Messaging" philosophy based on a decoupled architecture. Rather than sending complex bundled packets, each physical device acts as separate logical sensors, transmitting independent MQTT messages for each measurement type. This approach ensures the backend remains generic and can scale to new sensor types without structural changes.

Every sensor payload follows a strict JSON "Data Contract" to maintain consistency across the platform:

– **id:** A unique UUID for hardware identification.
– **kind:** The measurement type (e.g., `flow_rate`), which Node-RED maps directly to the InfluxDB measurement name.
– **units:** Saved as metadata (Tags) to differentiate between measurement scales.
– **data:** The raw numerical value used for real-time graphing and historical analysis.
– **time_stamp:** A Unix timestamp used to index data points accurately on the temporal X-axis.

3.2 Database Selection and Logic

InfluxDB was selected as the primary data store due to its native optimization for time-series streams. Unlike relational databases, it provides high-ingest performance for million-point streams and automated lifecycle management through Retention Policies (e.g., `KEEP 30 DAYS`), preventing storage exhaustion. Data ingestion is handled directly within Node-RED to allow for logic-first preprocessing, enabling the system to filter bad readings or outliers before they reach the database.

3.3 Implementation Details

The development of the prototype was executed through a hardware-software integration process, prioritizing modularity and real-time responsiveness. Figure 1 presents the general architecture of the proof-of-concept solution, including the data pipeline, from sensor information to ML predictions. The complete implementation details are categorized as follows.

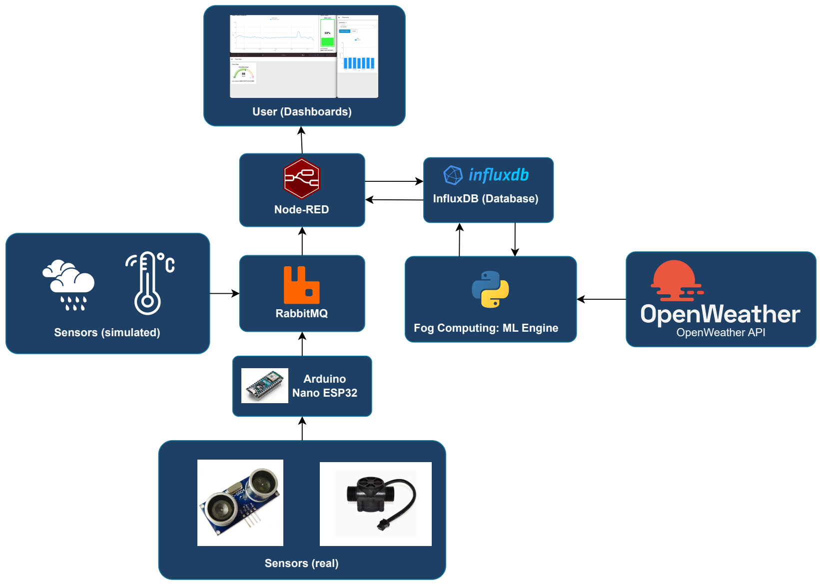

Fig. 1. Architecture of the prototype implemented. In comparison to the original architecture proposed for the application, we include simulated sensors and there is no actuation pipeline.

Hardware Setup and Data Acquisition The physical layer utilizes an Arduino Nano ESP32 units for local data acquisition. These controllers interface with a diverse sensor suite, including water flow sensors for monitoring inflow and outflow rates, rain gauges (pluviometers) for rainfall intensity, and pressure sensors that determine water levels via hydrostatic pressure. Additionally, power monitoring sensors track turbine voltage and current to calculate energy output. For the prototype, sensor data is simulated and scaled to mirror real-world hydrological conditions.

Communication Infrastructure Data transmission is built on a decoupled "Atomic Messaging" philosophy. Each sensor measurement (e.g., temperature, distance, flow) is sent as an independent MQTT message to its own specific topic (e.g., `sensors/temperature/ws_01`). While the final deployment is designed to scale to LoRaWAN due to its high range, the current prototype utilizes MQTT over Wi-Fi for efficient data transfer to the Raspberry Pi gateway.

Middleware and Backend Logic The Raspberry Pi serves as the central processing hub, running a Dockerized stack consisting of RabbitMQ, InfluxDB, and Node-RED.

– **Message Broker:** RabbitMQ handles the routing of MQTT messages, ensuring reliable and secure delivery through TLS encryption.

– **Data Storage:** InfluxDB was implemented for time-series storage, utilizing "Retention Policies" to automatically manage data lifecycles and prevent storage saturation.
– **Flow Engine:** Node-RED is the primary logic coordinator, responsible for filtering sensor readings and transforming JSON payloads into database-ready formats.

Predictive Analytics and Machine Learning To support proactive dam management, a machine learning model was developed to forecast reservoir water level evolution with a seven day horizon. The objective of this component is not full automation, but rather to provide predictive insights that support safer operational decisions within the Cyber-Physical System.

Initial Model As an initial approach, a Gradient Boosting Machine (GBM) model was developed in Python. The model was trained using historical hydrological data from the *Sistema Nacional de Informação de Recursos Hídricos* (SNIRH) [2] and integrated with meteorological data obtained from the OpenWeather API [4]. The system engineers features based on recent flow values and weather data to predict daily inflow and outflow trends.

Identified Limitations During validation and stress testing, limitations were observed in the predictive behaviour of the initial model. Although the model performed adequately under typical operating conditions, its predictions were strongly influenced by seasonal patterns present in the historical dataset.

The available data, spanning from 2023 to 2025, contained very few extreme drought events. As a result, the model showed limited sensitivity to prolonged high temperature and dry conditions, occasionally predicting stable or increasing water levels under simulated heatwave scenarios. This behaviour revealed a tendency towards seasonality overfitting, where calendar related correlations dominated over the physical drivers governing reservoir dynamics. In a safety critical context, this limitation reduces the reliability of drought risk anticipation.

Physics-Informed Feature Engineering To address this limitation, domain knowledge was explicitly incorporated into the learning process through feature engineering. A synthetic feature, referred to as the evaporation index, was introduced to represent environmental stress on the reservoir.

This index combines temperature, wind speed and relative humidity into a single variable that reflects the potential for evaporation-driven water loss. By providing the model with a direct physical signal, it becomes less dependent on implicit correlations and more sensitive to environmentally driven water loss mechanisms.

Synthetic Data Augmentation Strategy In addition to feature engineering, a controlled synthetic data augmentation strategy was applied during training. Synthetic samples were generated to represent prolonged drought scenarios characterized by high temperature, negligible rainfall and increased wind intensity.

For these synthetic observations, a gradual downward trend in water level was imposed, simulating daily water loss consistent with basic mass conservation principles. These synthetic samples were used exclusively during training and were combined with real historical data to form a hybrid dataset. This approach allows the model to learn physically plausible behaviour under extreme conditions that are absent from the original dataset, without replacing or distorting real observations.

Delta-Based Forecasting Strategy To further improve prediction stability and physical consistency, the forecasting formulation was revised. Instead of predicting absolute reservoir levels, the model was trained to predict daily variations, or deltas, in water level.

The final multi-step forecast is reconstructed by cumulatively summing the predicted deltas starting from the current observed level. This strategy ensures that predictions remain grounded in the system's present state and reduces the risk of unrealistic level jumps over longer horizons. It also aligns naturally with physical reasoning, as reservoir evolution is governed by incremental inflow and outflow processes.

Together, these improvements resulted in a physics-informed predictive model that demonstrates significantly improved behaviour under stress scenarios, while preserving accurate responses during normal operating conditions. This iterative refinement highlights the importance of embedding domain knowledge into data-driven models when operating under limited or imbalanced real-world datasets.

Human-Machine Interface (HMI) The system provides a comprehensive dashboard built in Node-RED for real-time visualization of all monitored variables. This interface includes automated alerts for overflow or drought conditions and provides manual override buttons for solenoid valve control, ensuring a "Human-in-the-Loop" safety mechanism.

3.4 Implementation Constraints and Design Deviations

During the transition from the theoretical system design to the physical prototype, several constraints necessitated deviations from the original proposed solution. These changes highlight the challenges of hardware availability and the adaptation required for small-scale modeling.

Water Level Sensing: Pressure vs. Ultrasonic Distance The initial design proposed a hydrostatic pressure transducer to measure water levels. However, for the current prototype implementation, an ultrasonic distance sensor was utilized as a primary alternative. While a pressure sensor measures the weight of the water column from the bottom, the ultrasonic sensor measures the distance from the top of the reservoir to the water surface. This shift required a modification in planing for the reservoir model. The ultrasonic sensor, is cost-effective and easier to implement in a small-scale setup but extremely complicated to apply to a real world scenario.

Flow Control: Manual vs. Automated Actuation The proposed architecture included a solenoid valve as an automated actuator to regulate water flow based on AI-driven commands from the Cognition layer. Due to hardware procurement limitations, the physical actuation is currently performed manually. To maintain the integrity of the Cyber-Physical loop, the system still generates "control recommendations" via the Node-RED dashboard. The operator acts as the physical link in the feedback loop, executing manual gate adjustments in response to the system's predictive alerts for overflow or drought.

4 Evaluation and Results

4.1 Testing Methodology

In order to assess the performance and effectiveness of the proposed system, a mockup model was developed Figure 2. This model intends to simulate real-world conditions and provide a controlled environment for testing various aspects of the system.

The model consisted of two boxes: 1) the dam reservoir (upper box) and 2) the downstream river (lower box). The reservoir box was filled with water to simulate different water levels, while the downstream river box was used to observe the effects of water release from the dam. The dam reservoir box had a hose that allowed water to flow into the downstream river box while being controlled by a water flow sensor. Our group intended to use a flow gate actuator to control the water flow, but unfortunately, it was not available. Therefore, the water flow was manually adjusted by inserting a piece on the hose to simulate the gate's partial opening and closing. The dam reservoir box also had an ultrasonic distance sensor to measure the water level, which provided real-time data for analysis. Due to water fluctuations, and consequent reflections, the ultrasonic distance sensor readings were quite noisy. A possible pratical solution to this problem would be to place a floating platform on the water surface to provide a more stable reflection point for the ultrasonic distance sensor. Both the water flow sensor and the ultrasonic distance sensor were connected to an Arduino board, placed in the exterior of the dam reservoir box, to collect and transmit data to a computer for further analysis.

This setup allowed us to simulate dam operations and gather data for evaluating the system's performance in a controlled environment.

4.2 Real-world application

Our system was designed to address challenges with dam systems management, by providing extra analytics to support decisions and prevent failures.

A primary concern in dam operations is the risk of overflow. To mitigate this the system uses multiple sensors, such as pluviometers to gain an accurate insight into the exact quantity of rain, compared to weather estimations, the area is receiving. Inflows from rivers can also be monitored through the use of a water flow sensor placed at the dam's main water entrance or if the only inflow source is another dam, the system can use flow measurements from the upstream's dam gate as an approximate value. This gives us valuable insight and allows the model to have an improved performance based on the quantity and sources of data and provides dam technicians with real-time, comprehensive insights into reservoir status.

For optimizing energy production, the system requires two types of sensors and one actuator: a water flow sensor to monitor outflow, a dynamo equipped with a sensor to measure generated electricity, and a flow gate actuator. This configuration enables remote and automated control of energy production, allowing

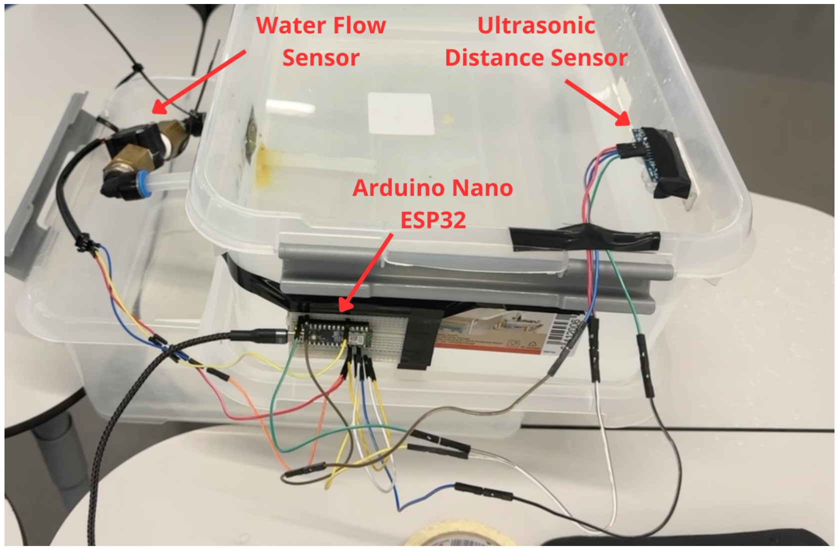

Fig. 2. Testing Model

the system to adjust operations dynamically in response to changing conditions or to be remotelly actuated by dam technicians.

This would allow our system to provide analytics for technicians to take informed decisions and a way to act on this information in an entirely remote manner helping fix another common issue in dams, their remote or hard to access locations.

4.3 Result Analysis

The full prototype functioned as expected, providing real-time data within the expected latency constraints. Unfortunately, the lack of access the a flow gate actuator made it impossible to extract accurate measures and evaluate our model extensively. As such, we'll focus this section on the details of the experimental setup, as well as how the sensor input was reflected in the UI.

The ultrasonic distance sensor used to measure the water level had a measurement error of ± 0.5 centimeters which, given the height of the box used, resulted in an fluctuation of approximately $\pm 7\%$ with regards to the current water capacity of the dam. This led to some impact in our inference results, as the inference model often received a drastic variation in the readings of that particular sensor. In a real-world system, this margin of error would be negligible, as dams are usually several dozen meters in height - 3 orders of magnitude above the measurement error of our instruments - meaning that the fluctuation in dam capacity would be considerably smaller. The retrieved results were consistent, except for spurious readings with a particularly large fluctuation, and the real-time updates ensured a dam operator could stay informed regarding the current state of the dam's used capacity. The dashboard worked as expected, providing valuable insights into the dam, complete with a functioning color code for fast visual identification.

Due to resource constraints we could not simulate different pluviosity conditions, nor different upstream inflows. While this could be circumvented using artificial data, we decided against it, as it would defeat the purpose of having a physical model to evaluate our system. Further work on this prototype would require simulating these conditions, as well as extreme cases - i.e floods, monsoons, emergencies in upstream dams - to ensure our model was resilient in these situations.

Given the displayed performance, specially with regards to latency in real-time updates, we consider our model adequately validates the feasibility of our solution. While a real implementation would require far more testing and validation than what we were able to perform, we managed to prove the functionality of the core concept, as well as validate telemetry and inference.

5 Evaluation and Results

5.1 Testing Methodology

Evaluation relies on the daily split spanning 1 January to 19 February 2024, with the learned XGBoost model trained on a broader historical window and the persistence baseline retaining each previous day's reading. Predictions are contrasted against the measured volumes to compute MAE, RMSE, MAPE and R^2 per model, ensuring that the performance comparison isolates the effect of the modeling strategy rather than dataset selection. This test window is representative of both stable periods and short-lived spikes, so the diagnostics incorporate both global error summaries and diagnostic plots to verify calibration.

5.2 Quantitative Analysis

The XGBoost model delivers clear gains: MAE drops from 0.215 to 0.150 and RMSE falls from 0.238 to 0.182, representing relative reductions of roughly 30% and 25%. MAPE decreases to 1.484% (from 2.143%), and R^2 improves from 0.882 to 0.930, approaching the variance-explained ceiling. These values appear in Table 2.

Model	**MAE**	**RMSE**	**MAPE (%)**	R^2
XGBoost	0.150	0.182	1.484	0.930
Persistence	0.215	0.238	2.143	0.882

Table 2. Aggregated metrics per model on the January–February 2024 test window.

The scatter plot in Figure 3 confirms that XGBoost predictions stay close to the diagonal, indicating largely unbiased forecasts. Residuals follow a symmetric distribution around zero (Figure 4) with a mean of 0.013, standard deviation of 0.18 and a maximum absolute deviation below 0.5 units, which shows that outliers are rare and that the model avoids severe overshooting of the true value.

5.3 Case Study: Dynamic Event

Figure 5 shows how the model responds to abrupt changes. Around 11 January, the observed volume jumps by 0.33 units in a single day. The learned model captures that increase almost immediately and keeps tracking the overall level in the following days, whereas persistence lags behind by simply repeating the previous day's value. This behaviour demonstrates that the regression effectively captures the physical displacement caused by meteorological inputs while preserving the reservoir's inertia.

The feature importances in Figure 6 confirm the delta-learning strategy: the model prioritizes the previous day's and afternoon's volumes plus afternoon temperature (in Celsius), followed by other lagged volumes, minimum temperature,

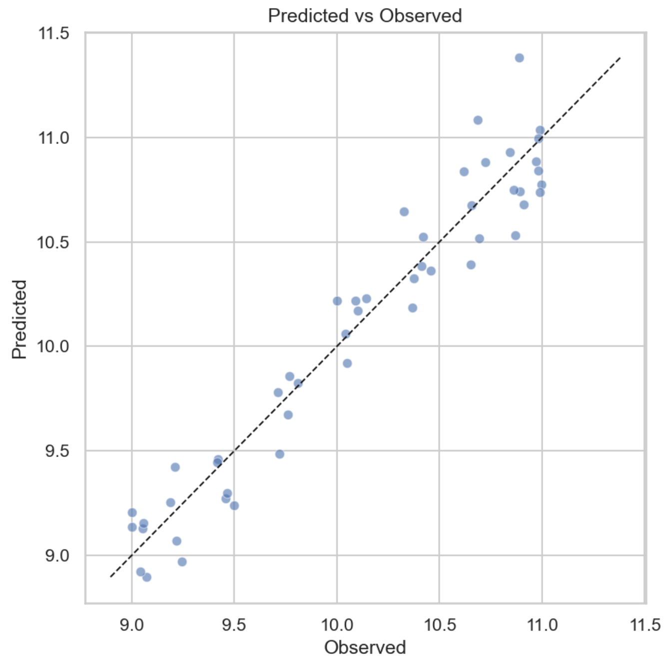

Fig. 3. Predicted versus observed values. The alignment with the diagonal illustrates the consistent accuracy of XGBoost.

and forecasted temperature/precipitation features. These physical signals help differentiate the response to the climate under critical moments while the baseline simply memorizes the last available reading.

5.4 Comparative Visualizations

Three additional visualizations illustrate robustness and sensitivity beyond the aggregate metrics:

- **Comparative Stress Test**: exposes the trajectory of the learned model, the previous version and the persistence baseline during an extreme drought scenario, including a threshold around 20% capacity. Figure 7 shows that the physics-aware model signals the drop much earlier while the legacy options remain on the wrong side of the threshold.
- **Physics Blindness Test**: plots the predicted daily change (delta) versus temperature for the two trained regressors. The physics-aware XGBoost adapts its prediction upward as heat increases, whereas the older model stays largely flat; this contrast is captured in Figure 8.
- **Scenario Battle**: compares the predicted deltas for both models under three critical situations (normal day, heatwave/drought, heavy storm). Figure 9

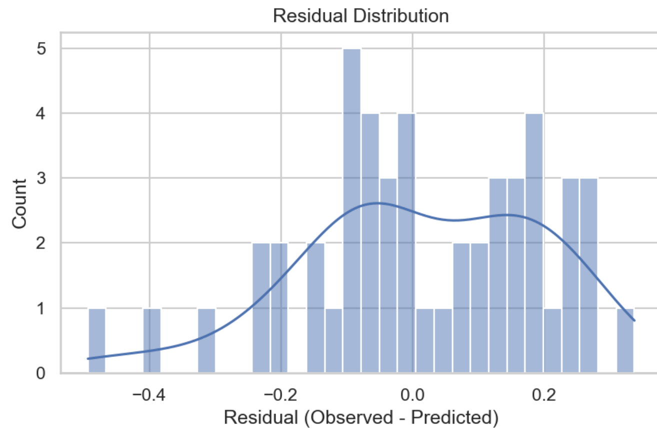

Fig. 4. Histogram of residuals computed from `predictions.csv`. Most errors cluster near zero, with little mass above 0.2 units.

makes it explicit that the new model adjusts per scenario (slowing inflow under heat, anticipating flooding under storms) while the older regression lacks such sensitivity.

These visualizations complement the average metrics described earlier and reinforce the "physics-aware" strategy: the expanded model family not only shrinks aggregated error but also reacts intelligently across stress-testing, temperature shifts and scenario-specific decision-making.

5.5 Limitations and Next Steps

The current experiment does not include risk classes or event-specific metrics, so the evaluation remains centered on global average errors. Future steps should define operating classes (dry, normal, overflow risk) and compute confusion matrices to quantify critical false negatives. We also need to replace persistence-based weather inputs with real forecast feeds and validate the model on other reservoirs to ensure generalization. Finally, automating the pipeline that refreshes `predictions.csv` and reruns `generate_plots.py` would keep the figures and tables cited in this section synchronized with new data.

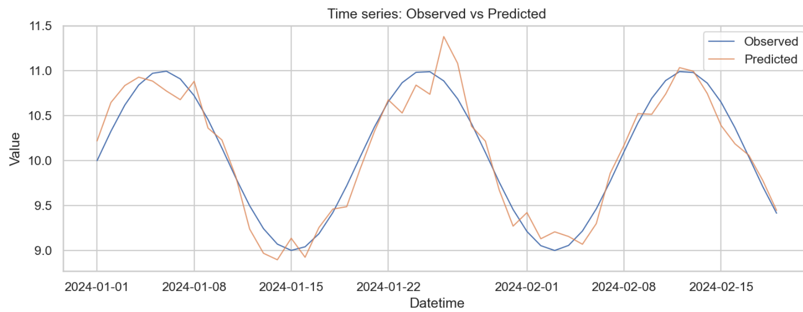

Fig. 5. Time series of observed and predicted values. The learned forecast closely follows the mid-January oscillation with minimal delay.

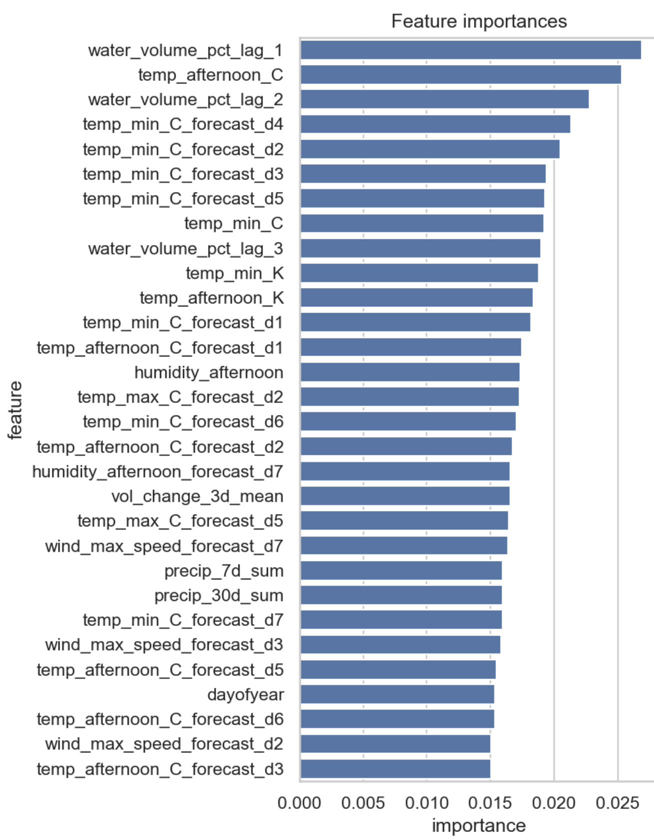

Fig. 6. Top feature importances for the XGBoost model. Volume lags and late-afternoon temperature lead, supporting the physics-informed delta learning heuristic.

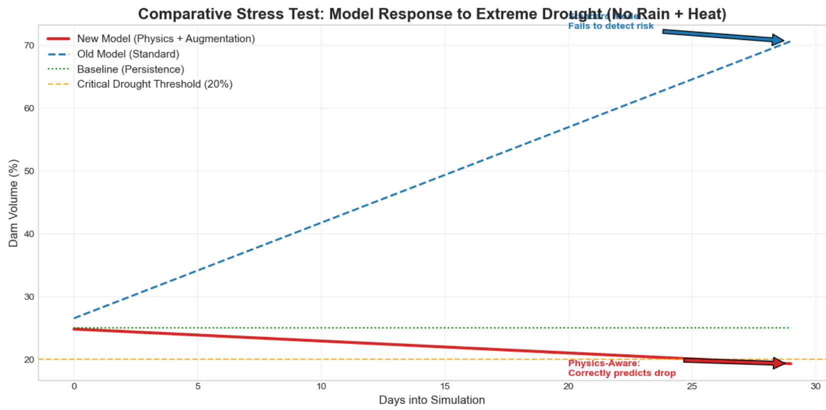

Fig. 7. Stress-test trajectory comparing predictions from the physics-aware model, the earlier standard model, and the persistence baseline under prolonged drought. The critical 20% threshold highlights the earlier warning provided by the new model.

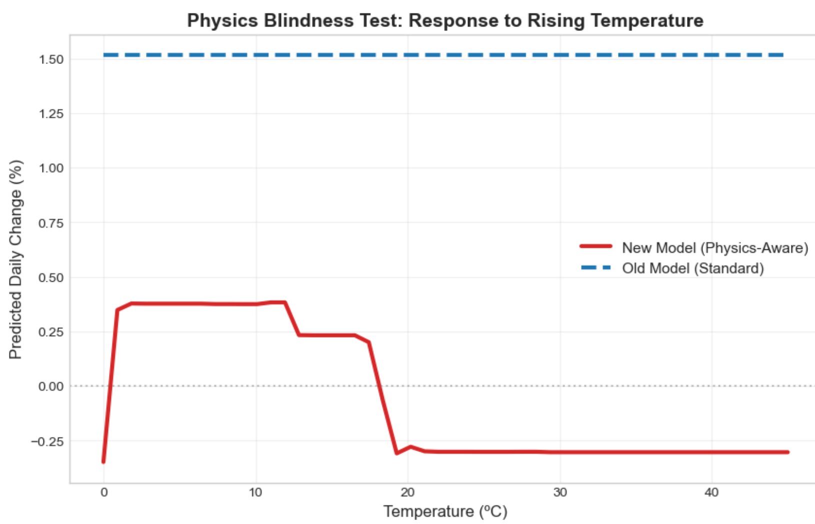

Fig. 8. Physics Blindness Test: the delta predicted by each model as temperature increases. The new model reacts to the rising heat (physics-aware behavior) while the older model remains mostly flat.

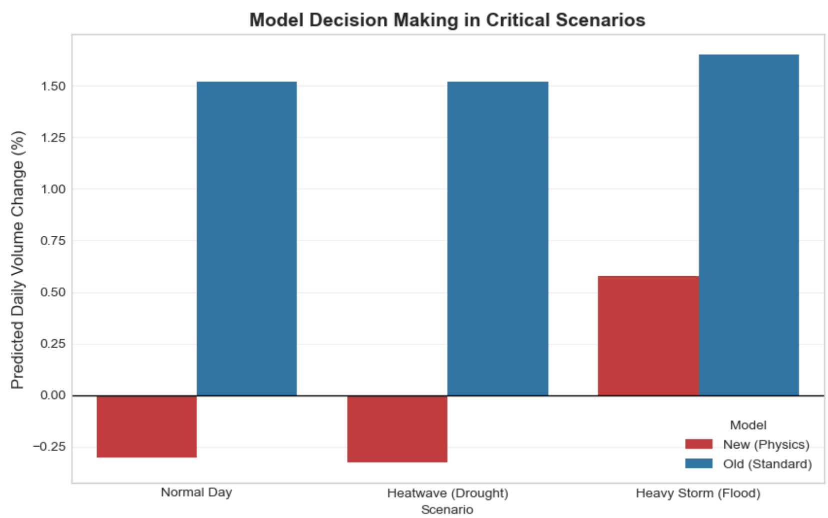

Fig. 9. Scenario Battle: predicted daily changes for three critical scenarios and for both model versions. The physics-aware model adjusts predictions per scenario, whereas the legacy model is less responsive.

6 Discussion

This section presents a discussion of the benefits and limitations of the proposed solution, including the prototype and the ML predictive analytics.

6.1 Prototype Discussion

Although the prototype successfully demonstrated the feasibility of a unified IoT monitoring stack, it also includes several limitations.

A simplification of the prototype was the use of an ultrasonic sensor. It was a good alternative for small-scale testing on a controlled environment, though it remains sensitive to surface turbulence and even more susceptible to noise from particles in the air compared to hydrostatic methods. Furthermore, while the current manual control of the dam gate is a limitation of the physical prototype, it can validate other aspects of the solution, such as monitoring sensor information and the human interface for the operator. This ensures that even without direct electronic actuation, the decision support system provides actionable intelligence that enhances operational safety. Future iterations can focus on integrating solenoid actuators to achieve the full "Self-Adaptation" envisioned in the 5C CPS architecture.

In addition, while the predictive analytics module enhances decision support, it does not directly suggest actions to take. Consequently, the human operator needs to manually translate the insights into specific control actions.

6.2 Machine Learning Discussion

This subsection discusses the main design decisions, benefits, and limitations of the machine learning component integrated into the proposed CPS architecture. The discussion focuses on the trade-offs between predictive performance and system safety, as well as on the strategies adopted to reduce bias and improve robustness in safety-critical operating scenarios.

Initial Approach and Identified Limitations The initial machine learning approach aimed to capture historical water level behaviour, with a particular focus on seasonal patterns. Although this baseline model achieved reasonable predictive performance on past data, it revealed important limitations. In particular, it showed a strong bias towards seasonality and limited sensitivity to rare but safety-critical situations, such as extreme low water level conditions associated with dry risk events, since these events were underrepresented in the training dataset.

Safety-Oriented Design Decisions Within the context of Cyber-Physical Systems and IoT, prediction errors may have direct physical consequences. In dam management scenarios, underestimating critical conditions can compromise operational safety and, in extreme cases, lead to catastrophic outcomes. For this reason, the machine learning component was deliberately redesigned following a safety-oriented approach, prioritizing robustness and conservative behaviour rather than purely maximizing predictive accuracy.

Synthetic Data and Safety-Performance Trade-off One of the main challenges identified was the limited availability of real-world data representing rare and low-frequency events. To address this issue, synthetic data generation techniques were used to enrich the training dataset with safety-critical scenarios. This strategy reduced the model's dependence on dominant seasonal patterns and increased its sensitivity to dry risk conditions. As a result, the model adopted a more cautious predictive behaviour, supporting safer decision-making within the CPS. As expected, this design choice introduced a trade-off between system safety and predictive performance. By explicitly modeling rare events during training, the system becomes better prepared to support operators in high-risk situations, even when historical data alone is insufficient to fully capture such conditions.

Interpretation of Performance Degradation A degradation in predictive performance was observed when evaluating the model using the available test dataset. This outcome was anticipated, since the test data originates from the same dam and therefore follows seasonal dynamics similar to those present in the original training data. By reducing its reliance on these seasonal correlations, the model sacrifices performance on familiar patterns in exchange for improved robustness under critical operating conditions. From a CPS perspective, this behaviour is desirable, as it reduces the likelihood of overly optimistic predictions in scenarios characterized by uncertainty and environmental variability.

Alignment with CPS and IoT Principles The observed trade-offs reflect a conscious engineering decision aligned with core CPS and IoT principles, where safety, reliability, and risk mitigation take precedence over maximum statistical performance. In safety-critical infrastructures, conservative predictions are often preferable to highly optimized models that may fail under rare but severe conditions.

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